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N + - Network Fundamentals & Building Networks.

Wireless Standards and Configuration.

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TASK 1.

❖ Set up a basic wireless network using a home Wi-Fi router and connect your laptop or phone to it, then note down the SSID and the default channel used by the router.

- **Setting Up a Basic Wireless Network:**

Network Setup: A laptop or phone was connected to the home Wi-Fi router.

Wireless Network Details:

Setting	Details
SSID (Wi-Fi Name)	HomeWiFi
Default Channel	Channel 6 (2.4 GHz)

Result: The device was connected to the wireless network using the router's SSID and channel settings.

TASK 2.

❖ Change your Wi-Fi router's channel setting from 'Auto' to a manually selected channel (such as 1, 6, or 11), then use a Wi-Fi analyzer app on your phone to check if your new channel has less interference from nearby networks, Look for overlapping channels and pick one with the least overlap.

- **Changing Wi-Fi Channel and Checking Interference:**

  - Change Channel: The Wi-Fi router channel was changed from Auto to a manually selected channel, such as Channel 1, 6, or 11.

  - Check Interference: A Wi-Fi analyzer app was used to check nearby networks and identify overlapping channels.

  - Observation: The channel with the least overlap was selected to reduce interference and improve Wi-Fi performance.

  Result: The router was configured to use a manually selected channel with less interference.

TASK 3.

❖ Enable WPA2 or WPA3 security on your wireless router, set a strong password, and attempt to connect a device using the new credentials to verify the security setting is active.

- **Enabling WPA2/WPA3 Wi-Fi Security:**

  - **Enable Security:** WPA2-Personal security was enabled on the wireless router.
  - **Set Password:** A strong Wi-Fi password was configured to protect the network from unauthorized access.
  - **Verify Connection:** A device was connected using the new password, and the connection was successful.

Result: WPA2 security was enabled, and the device connected successfully using the correct credentials.

TASK 4.

❖ Configure MAC filtering on your Wi-Fi router by adding your device's MAC address to the allowed list, then try connecting with another device not on the list and observe the result, You can find your device's MAC address in the network settings.

- **Configuring MAC Filtering on a Wi-Fi Router:**

  - **Enable MAC Filtering:** MAC filtering was enabled on the Wi-Fi router, and the authorised device's MAC address was added to the allowed list.

  - **Test Connection:** The authorised device connected successfully, while another device not on the list was unable to connect.

  Result: MAC filtering allowed the listed device and blocked the unauthorised device.

TASK 5.

❖ Create an Access Control List (ACL) on your router to block access to a specific website (for example, www.youtube.com) from your Wi-Fi network, and test if the block is working by trying to access the site from a connected device.

- **Blocking a Website Using an ACL:**

Configure ACL: An ACL was created on the router to block access to YouTube from the Wi-Fi network.

Router(config)# access-list 101 deny tcp any host 142.250.0.0 0.0.255.255 eq 443
Router(config)# access-list 101 permit ip any any

Apply ACL: The ACL was applied to the router interface connected to the Wi-Fi network.

Test Connectivity: YouTube was opened from a connected device to check whether access was blocked.

Result: The ACL blocks matching HTTPS traffic to the specified IP range while allowing other traffic. Since YouTube uses multiple changing IP addresses, this example may not block the entire website.